

FALL 2025-2026

BMAT205L - DISCRETE MATHEMATICS AND GRAPH THEORY

Module-3 : Tutorial -5

(To access the students' understanding on Computing Techniques and Recurrence Relations)
11 September, 2025

1. In a class of 50 students, 20 students play football, and 16 students play hockey. It is found that 10 students play both the games. Find the number of students who play neither.
2. How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school?
3. Find the number of bit strings of length 10 that either begin with 1 or end with 0.
4. The chairs of an auditorium are to be labeled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently?
5. In a survey of 100 students, it was found that 40 studied Mathematics, 64 studies Physics, 35 studied Chemistry, 1 studied all the three subjects, 25 studied Mathematics and Physics, 3 studied Mathematics and Chemistry, 20 studied Physics and Chemistry. Use the principle of inclusion and exclusion to find the number of students who studied Chemistry only and the number of students who studied none of these subjects.
6. If 200 faculty members can speak French and 50 can speak Russian while 20 can speak both French and Russian, find how many faculty members can speak either French or Russian.
7. How many permutations can be made out of the letters of the word 'BASIC'? How many of these (i) begin with B? (ii) End with C? (iii) B and C occupy the end places?
8. In how many ways a football team of eleven players can be chosen out of 17 players, when (i) five particular players are to be always included (ii) two particular players are to be always excluded.
9. Find the number of integers between 1 to 100 that are not divisible by any of the integers 2,3,5 or 7.
10. How many positive integers not exceeding 1000 are divisible by none of 3,7 and 11?
11. Use generating functions to solve the following recurrence relations:
 - (i) $a_n - 2a_{n-1} - 3a_{n-2} = 0, n \geq 2$ with $a_0 = 3, a_1 = 1$
 - (ii) $T(k) - 7T(k-1) + 10T(k-2) = 6 + 8k, T(0) = 1, T(1) = 2$
 - (iii) $y_{n+2} - 4y_{n+1} + 3y_n = 0, y_0 = 2, y_1 = 4$
 - (iv) $a_n = 8a_{n-1} + 10^{n-1}, a_0 = 1$
 - (v) $a_n = 8a_{n-1} - 16a_{n-2}, n \geq 2, a_0 = 16, a_1 = 80$
 - (vi) $a_n = 2a_{n-1} + 2^n, a_0 = 2$
 - (vii) $a_n = 3a_{n-1} + 1, n \geq 1, a_0 = 1$
 - (viii) $s(n+2) - 5s(n+1) + 6s(n) = 0, n \geq 0, s(0) = 1, s(1) = 1$
 - (ix) $a_{n+2} - 2a_{n+1} + a_n = 2^n, n \geq 0, a_0 = 2 \text{ and } a_1 = 1$